

Phase I/II Operating Characteristics

Continuous enrollment | 56-day arrival | IPDE alpha = 0.6

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	9.4	4.6	2.3	0.9	0.3	82.5
C2 alpha=0.6 one-stage	9.2	2.5	0.7	0.1	0	87.5
C3 alpha=0.6 one-stage	8.5	2.6	0.2	0	0	88.7
Oracle C1 alpha=0 two-stage	9.2	5.1	2.2	0.8	0.3	82.4
C2 alpha=0.6 two-stage	9.7	2.5	0.6	0.1	0	87.1
C3 alpha=0.6 two-stage	9.1	2.7	0.2	0	0	88
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	11	4.2	2.3	0.9	0.2	
C2 alpha=0.6 one-stage	11.7	3.3	1.3	0.3	0.1	
C3 alpha=0.6 one-stage	12	3	1	0.2	0	
Oracle C1 alpha=0 two-stage	10.9	4.3	2.4	0.9	0.3	
C2 alpha=0.6 two-stage	11.6	3.4	1.4	0.3	0.1	
C3 alpha=0.6 two-stage	11.9	3.1	1	0.2	0	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	23.9	21	19.3	21.8	13.1	0.9
C2 alpha=0.6 one-stage	23.6	26.9	19.9	17.6	11.8	0.2
C3 alpha=0.6 one-stage	21.2	28.4	22.6	17.8	9.9	0.1
Oracle C1 alpha=0 two-stage	24.1	21.9	18.3	21.1	13.7	0.9
C2 alpha=0.6 two-stage	23.4	28.9	17.2	19.4	10.9	0.2
C3 alpha=0.6 two-stage	22.3	28.9	22.6	15.9	10.1	0.2
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	7.2	6.7	5.8	5.8	4.5	
C2 alpha=0.6 one-stage	7.3	7.5	5.8	5.4	4.1	
C3 alpha=0.6 one-stage	7.1	7.5	6.3	5.3	3.8	
Oracle C1 alpha=0 two-stage	7.1	6.7	5.7	5.8	4.8	
C2 alpha=0.6 two-stage	6.8	7.4	5.7	5.7	4.4	
C3 alpha=0.6 two-stage	6.9	7.3	6.1	5.5	4.2	

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Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	4.9	47	28.4	12.3	2.2	5.2
C2 alpha=0.6 one-stage	6.3	51.6	27.1	8.2	0.9	5.9
C3 alpha=0.6 one-stage	5.4	52.4	29.1	7	0.5	5.7
Oracle C1 alpha=0 two-stage	4.5	48.5	29.1	10.9	1.9	5.1
C2 alpha=0.6 two-stage	6.1	54	26.4	7.2	0.7	5.6
C3 alpha=0.6 two-stage	5.8	53	29.5	5.1	0.4	6.3
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	5.2	11.3	7.6	4.1	1.3	
C2 alpha=0.6 one-stage	5.6	12	7.5	3.5	0.7	
C3 alpha=0.6 one-stage	5.6	12.3	7.7	3	0.6	
Oracle C1 alpha=0 two-stage	5.1	10.7	8	4.3	1.4	
C2 alpha=0.6 two-stage	5.4	11.4	7.8	3.8	1	
C3 alpha=0.6 two-stage	5.5	11.5	8.1	3.4	0.9	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	0.8	1.4	6	59.4	32.4	0
C2 alpha=0.6 one-stage	0.4	0.7	3.5	63.8	31.5	0
C3 alpha=0.6 one-stage	0.4	0.9	3.2	61.5	33.9	0
Oracle C1 alpha=0 two-stage	1.1	1.2	6.3	56	35.4	0
C2 alpha=0.6 two-stage	0.6	0.9	4.2	57.5	36.8	0
C3 alpha=0.6 two-stage	1	1.1	4.3	53.4	40.1	0
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	3.4	3.5	4.5	12.5	6.2	
C2 alpha=0.6 one-stage	3.4	3.4	4.2	13.5	5.5	
C3 alpha=0.6 one-stage	3.4	3.4	4.3	13	6	
Oracle C1 alpha=0 two-stage	3.4	3.4	4.3	10.9	8	
C2 alpha=0.6 two-stage	3.4	3.3	4	11.3	8	
C3 alpha=0.6 two-stage	3.4	3.3	4.1	10.6	8.7	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	10.7	25.3	44.6	9.5	0.6	9.3
C2 alpha=0.6 one-stage	11.8	28.5	45.7	5.5	0.2	8.3
C3 alpha=0.6 one-stage	14.4	30.2	41.8	5.2	0.3	8.1
Oracle C1 alpha=0 two-stage	11.3	23.8	46.3	9	0.9	8.7
C2 alpha=0.6 two-stage	12.5	30.8	42.5	6	0.1	8.1
C3 alpha=0.6 two-stage	13.4	31.6	41.9	4.3	0.2	8.5
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	7.3	8	9.8	3.4	0.6	
C2 alpha=0.6 one-stage	7.8	8.8	9.6	2.5	0.3	
C3 alpha=0.6 one-stage	8.1	9.1	9.5	2.1	0.2	
Oracle C1 alpha=0 two-stage	6.9	7.9	9.7	3.8	0.9	
C2 alpha=0.6 two-stage	7.5	8.7	9.2	3.1	0.6	
C3 alpha=0.6 two-stage	7.8	8.9	9.2	2.7	0.5	
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	3	6.5	8.7	11.8	65.3	4.7
C2 alpha=0.6 one-stage	5	8.9	10.9	15.4	54.7	5.1
C3 alpha=0.6 one-stage	6.1	8.1	13.6	17.4	49.6	5.3
Oracle C1 alpha=0 two-stage	4.6	7.2	8.6	11.6	63.2	4.8
C2 alpha=0.6 two-stage	5.6	9.5	10.5	15.4	54.6	4.4
C3 alpha=0.6 two-stage	6.1	10.1	13.3	16.1	49.7	4.7
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	4.6	4.6	5	5.5	10.2	
C2 alpha=0.6 one-stage	4.8	5	5.4	5.7	8.9	
C3 alpha=0.6 one-stage	4.8	5.1	5.9	5.4	8.4	
Oracle C1 alpha=0 two-stage	4.5	4.5	4.9	5.6	10.3	
C2 alpha=0.6 two-stage	4.7	4.9	5.3	5.8	9.1	
C3 alpha=0.6 two-stage	4.7	5.1	5.7	5.6	8.5	

Mean Unique Patients by Method

Mean number of unique patients per simulated trial

Scenario	1	16	20	24	27	38	Mean
Oracle C1 alpha=0 one-stage	18.7	30	29.5	30	29.1	29.8	27.8
C2 alpha=0.6 one-stage	10.9	18.2	18.6	19.6	19	18.6	17.4
C3 alpha=0.6 one-stage	9.3	15.1	16	17.3	16.5	15.5	15
Oracle C1 alpha=0 two-stage	18.8	30	29.5	30	29.1	29.8	27.9
C2 alpha=0.6 two-stage	10.9	18.2	18.7	19.7	19.1	18.6	17.5
C3 alpha=0.6 two-stage	9.4	15.2	16.2	17.6	16.8	15.6	15.1

Values are reported from Total_Unique_Patients; comparator counts without patient IDs are sums of dose-level means.

Rows are one-stage first, then two-stage. C1 is fixed at alpha=0; C2/C3 use alpha=0.6.